

Trigonometric Equations

Question1

If $\tan\left(\frac{\pi}{4} + \alpha\right) = \tan^3\left(\frac{\pi}{4} + \beta\right)$, then $\tan(\alpha + \beta) \cot(\alpha - \beta) =$

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Options:

A.

$$\sec^2 2\beta + \tan^2 2\beta$$

B.

$$\operatorname{cosec}^2 2\beta + \cot^2 2\beta$$

C.

$$2(\sec^2 2\beta + \tan^2 2\beta)$$

D.

$$4(\sec^2 2\beta + \tan^2 2\beta)$$

Answer: C

Solution:

$$\tan\left(\frac{\pi}{4} + \alpha\right) = \tan^3\left(\frac{\pi}{4} + \beta\right)$$

$$\text{let } x = \frac{\pi}{4} + \alpha, y = \frac{\pi}{4} + \beta$$

$$\therefore \tan x = \tan^3 y$$

$$\text{Now, } \alpha + \beta = x + y - \frac{\pi}{2}$$

$$\text{and } \alpha - \beta = x - y$$

$$\text{then, } \tan(\alpha + \beta) = \tan\left(x + y - \frac{\pi}{2}\right)$$

$$= -\cot(x+y)$$

$$\text{and } \cot(\alpha - \beta) = \cot(x - y)$$

$$\text{Thus, } \tan(\alpha + \beta) \cot(\alpha - \beta)$$

$$= -\cot(x+y) \cdot \cot(x-y)$$

$$\text{and } \cot(x+y) \cdot \cot(x-y)$$

$$\begin{aligned} &= \frac{(1 - \tan^4 y)(1 + \tan^4 y)}{\tan^2 y (1 + \tan^2 y)(1 - \tan^2 y)} \\ &= \frac{1 + \tan^4 y}{\tan^2 y} = \cot^2 y + \tan^2 y \end{aligned}$$

$$\text{Thus, } -\cot(\alpha - \beta) \tan(\alpha + \beta)$$

$$= -(\cot^2 y + \tan^2 y)$$

$$\text{and } \tan y = \tan\left(\frac{\pi}{4} + \beta\right) = \frac{1 + \tan \beta}{1 - \tan \beta}$$

$$\Rightarrow \cot y = \frac{1 - \tan \beta}{1 + \tan \beta}$$

Therefore,

$$\begin{aligned} \tan^2 y + \cot^2 y &= \frac{(1 - \tan \beta)^4 + (1 + \tan \beta)^4}{(1 - \tan^2 \beta)^2} \\ &= \frac{(1 - \tan \beta)^4}{(1 - \tan \beta)^2(1 + \tan \beta)^2} + \frac{(1 + \tan \beta)^4}{(1 - \tan \beta)^2(1 + \tan \beta)^2} \\ &= \left(\frac{1 - \tan \beta}{1 + \tan \beta}\right)^2 + \left(\frac{1 + \tan \beta}{1 - \tan \beta}\right)^2 \\ &= \frac{(1 + \tan^2 \beta - 2 \tan \beta)^2 + (1 + \tan^2 \beta + 2 \tan \beta)^2}{(1 - \tan^2 \beta)^2} \\ &= \frac{2(1 + \tan^4 \beta + 4 \tan^2 \beta) + 2(\tan^2 \beta + 2 \tan^3 \beta + 2 \tan \beta + \tan^2 \beta - 2 \tan^3 \beta - 2 \tan \beta)}{(1 - \tan^2 \beta)^2} \\ &= 2 \left[\frac{1 + \tan^4 \beta + 4 \tan^2 \beta + 2 \tan^2 \beta}{(1 - \tan^2 \beta)^2} \right] \\ &= 2 \left[\frac{(\tan^4 \beta + 2 \tan^2 \beta + 1) + (2 \tan \beta)^2}{(1 - \tan^2 \beta)^2} \right] \\ &= 2(\sec^2 2\beta + \tan^2 2\beta) \end{aligned}$$

Question2

If $0 \leq x \leq 3$ and $0 \leq y \leq 3$, then the number of solutions (x, y) of the equation $\left(\sqrt{\sin^2 x - \sin x + \frac{1}{2}}\right) 2^{\sec^2 y} = 1$ is

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Options:

A.

5

B.

2

C.

6

D.

1

Answer: B

Solution:

$$\text{Let } A = \sqrt{\sin^2 x - \sin x + \frac{1}{2}}$$

$$\text{and } B = 2^{\sec^2 y}$$

$$\text{then, } A \cdot B = 1$$

for A to be real

$$\sin^2 x - \sin x + 1/2 \geq 0$$

$$\text{but minimum value of } \sin^2 x - \sin x + 1/2 = 1/4$$

$$\text{at } x = \pi/6$$

$$\Rightarrow \sqrt{\sin^2 x - \sin x + \frac{1}{2}} \geq 1/2$$

$$\because x \in [0, 3] \Rightarrow x = \pi/6, 5\pi/6$$

$$\therefore B = 2^{\sec^2 y} \leq 2$$

$$\text{because } \sec^2 y = \frac{1}{\cos^2 y}$$

$$\Rightarrow |\cos x| \leq 1$$

$$\text{and } \sec^2 y \geq 1$$

So, $B = 2$ and $A = 1/2$ is only possible solution

$$\text{for } B = 2 \sec^2 y = 1 \text{ in } y \in [0, 3]$$

$$\sec^2 y = 1 \text{ at } y = 0$$

therefore, possible pairs are

$$(\pi/6, 0) \text{ and } (5\pi/6, 0)$$

$\therefore$ Number of solutions are 2 .

Question3

The number of solutions of the equation $4 \cos 2\theta \cos 3\theta = \sec \theta$ in the interval $[0, 2\pi]$ is

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Options:

A.

12

B.

8

C.

16

D.

4

Answer: A

Solution:

Given,

$$4 \cos 2\theta \cos 3\theta = \sec \theta$$

$$\Rightarrow 4 \cos \theta \cos 2\theta \cdot \cos 3\theta = 1$$

$$\Rightarrow 2 \cos 2\theta (\cos 4\theta + \cos 2\theta) = 1$$

$$\Rightarrow 2 \cos 2\theta \cos 4\theta + 2 \cos^2 2\theta = 1$$

$$\Rightarrow 2 \cos 2\theta \cos 4\theta + 2 \cos^2 2\theta - 1 = 0$$

$$\Rightarrow 2 \cos 2\theta \cos 4\theta + \cos 4\theta = 0$$

$$\Rightarrow \cos 4\theta (2 \cos 2\theta + 1) = 0$$

$$\text{either } \cos 4\theta = 0$$

$$\Rightarrow 4\theta = (2n+1)\frac{\pi}{2} \Rightarrow \theta = (2n+1)\frac{\pi}{8}$$

$$\Rightarrow \cos 2\theta = -\frac{1}{2} \Rightarrow \cos 2\theta = \cos \frac{2\pi}{3}$$

$$\Rightarrow 2\theta = 2n\pi \pm \frac{2\pi}{3} \Rightarrow \theta = n\pi \pm \frac{\pi}{3}$$

$$\because \theta \in (0, 2\pi)$$

$$\therefore \theta = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8} \rightarrow 8 \text{ solutions}$$

$$\text{and } \theta = \frac{\pi}{3}, \pi \pm \frac{\pi}{3}, 2\pi - \frac{\pi}{3} \rightarrow 4 \text{ solutions}$$

$$\therefore \text{Total number of solution} = 12$$

Question4

$$\tan \frac{2\pi}{7} \cdot \tan \frac{4\pi}{7} + \tan \frac{4\pi}{7} \cdot \tan \frac{\pi}{7} + \tan \frac{\pi}{7} \cdot \tan \frac{2\pi}{7} =$$

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Options:

A.

7

B.

-7

C.

3

D.

-3

Answer: B

Solution:

Given, expression is

$$\tan\left(\frac{2\pi}{7}\right) \cdot \tan\left(\frac{4\pi}{7}\right) + \tan\left(\frac{4\pi}{7}\right) \cdot \tan\left(\frac{\pi}{7}\right) + \tan\left(\frac{\pi}{7}\right) \cdot \tan\left(\frac{2\pi}{7}\right)$$

$$\text{Let } x = \frac{\pi}{7}, y = \frac{2\pi}{7} \text{ and } z = \frac{4\pi}{7}$$

So, the expression becomes

$$\tan(y) \cdot \tan(z) + \tan(z) \cdot \tan(x) + \tan(x) \cdot \tan(y)$$

$$\text{Now, } x + y + z = \frac{\pi}{7} + \frac{2\pi}{7} + \frac{4\pi}{7} = \frac{7\pi}{7} = \pi$$

Since, $x + y + z = \pi$, we can use the identity

$$\tan(x) \cdot \tan(y) + \tan(y) \cdot \tan(z) + \tan(z) \cdot \tan(x) = -7$$

$$\therefore \tan\left(\frac{\pi}{7}\right) \tan\left(\frac{2\pi}{7}\right) + \tan\left(\frac{4\pi}{7}\right) \cdot \tan\left(\frac{2\pi}{7}\right) + \tan\left(\frac{4\pi}{7}\right) \tan\left(\frac{\pi}{7}\right) = -7$$

Question5

The sum of the solutions of $\cos x \sqrt{16 \sin^2 x} = 1$ in $(0, 2\pi)$ is

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Options:

A.

$$2\pi$$

B.

$$\frac{13\pi}{2}$$

C.

$$\frac{17\pi}{4}$$

D.

$$4\pi$$

Answer: A

Solution:

Given, $\cos x \sqrt{16 \sin^2 x} = 1$

Since, $16 \sin^2 x = 4|\sin x|$

So, $\cos x(4|\sin x|) = 1$

$\Rightarrow 2 \sin 2x = \pm 1$

Also, $\sin 2x = \pm \frac{1}{2}$

$$x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}$$

$$\therefore \text{Sum} = \frac{\pi}{12} + \frac{5\pi}{12} + \frac{7\pi}{12} + \frac{11\pi}{12} = 2\pi$$

Question6

If $\sqrt{3} \cos \theta + \sin \theta > 0$, then

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Options:

A.

$$-\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

B.

$$-\frac{\pi}{3} < \theta < \frac{2\pi}{3}$$

C.

$$-\frac{2\pi}{3} < \theta < \frac{\pi}{3}$$

D.

$$-\frac{\pi}{6} < \theta < \frac{5\pi}{6}$$

Answer: B

Solution:

$$\begin{aligned}
& \sqrt{3} \cos \theta + \sin \theta > 0 \\
\Rightarrow & \frac{\sqrt{3}}{2} \cos \theta + \frac{1}{2} \sin \theta > 0 \\
\Rightarrow & \cos \frac{\pi}{6} \cos \theta + \sin \frac{\pi}{6} \sin \theta > 0 \\
\Rightarrow & \cos \left(\theta - \frac{\pi}{6} \right) > 0 \\
\therefore & \cos x > 0 \\
\Rightarrow & x \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right) \\
\therefore & -\frac{\pi}{2} < \theta - \frac{\pi}{6} < \frac{\pi}{2} \\
\Rightarrow & -\frac{\pi}{2} + \frac{\pi}{6} < \theta < \frac{\pi}{2} + \frac{\pi}{6} \\
\Rightarrow & -\frac{\pi}{3} < \theta < \frac{2\pi}{3}
\end{aligned}$$

Hence, $-\frac{\pi}{3} < \theta < \frac{2\pi}{3}$

Question7

The general solution satisfying both the equations $\sin x = -\frac{3}{5}$ and $\cos x = -\frac{4}{5}$ is

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Options:

A.

$$x = (2n + 1)\pi + \tan^{-1} \left(\frac{3}{4} \right), n \in Z$$

B.

$$x = 2n\pi + \tan^{-1} \left(\frac{3}{4} \right), n \in Z$$

C.

$$x = n\pi + \tan^{-1} \left(\frac{3}{4} \right), n \in Z$$

D.

$$x = n\pi \pm \tan^{-1} \left(\frac{3}{4} \right), n \in Z$$

Answer: A

Solution:

$$\because \sin x = -3/5 \text{ and } \cos x = -4/5$$

$\sin x$ and $\cos x$ both are negative

So, x must be lie in 3rd quadrant

Therefore,

$$x = (2n + 1)\pi + \theta$$

$$\tan \theta = \frac{(-3/5)}{(-4/5)} = 3/4$$

$$\Rightarrow \theta = \tan^{-1}(3/4)$$

$$\text{Hence, } x = (2n + 1)\pi + \tan^{-1}(3/4), n \in \mathbb{Z}$$

Question8

The number of solutions of the equation $\sec x \cdot \cos 5x + 1 = 0$ in the interval $[0, 2\pi]$ is

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Options:

A.

5

B.

8

C.

10

D.

12

Answer: C

Solution:

Given, trigonometric equation is

$$\sec x \cdot \cos 5x + 1 = 0$$

$$\Rightarrow \frac{\cos 5x + \cos x}{\cos x} = 0$$

$$\Rightarrow 2 \cdot \cos 3x \cdot \cos 2x = 0$$

$$\Rightarrow \cos 3x = 0 \text{ or } \cos 2x = 0$$

$$\Rightarrow 3x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}, \frac{11\pi}{2}$$

$(\because 0 \leq x \leq 2\pi)$
 $(\therefore 0 \leq 3x \leq 6\pi)$

$$\Rightarrow x = \frac{\pi}{6}, \frac{3\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{9\pi}{6}, \frac{11\pi}{6}$$

$$\text{Also, } 2x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2} \left(\because 0 \leq x \leq 2\pi \right)$$

$(\therefore 0 \leq 2x \leq 4\pi)$

$$\Rightarrow x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

$$\therefore \text{Total number of solutions} = 10$$

Question9

If $2 \sin x - \cos 2x = 1$, then $(3 - 2 \sin^2 x) =$

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Options:

A.

$$\sqrt{3}$$

B.

$$-\sqrt{3}$$

C.

$$\sqrt{5}$$

D.

$$-\sqrt{5}$$

Answer: C

Solution:

$$\begin{aligned}2 \sin x - \cos 2x &= 1 \\ \Rightarrow 2 \sin x - (1 - 2 \sin^2 x) &= 1 \\ \Rightarrow 2 \sin x - 1 + 2 \sin^2 x &= 1 \\ \Rightarrow 2 \sin^2 x + 2 \sin x - 2 &= 0 \\ \Rightarrow \sin^2 x + \sin x - 1 &= 0 \\ \Rightarrow \sin x &= \frac{-1 \pm \sqrt{1+4}}{2 \times 1} \\ \Rightarrow \sin x &= \frac{-1 \pm \sqrt{5}}{2} \\ \Rightarrow \sin x &= \frac{\sqrt{5}-1}{2} \\ \Rightarrow 3 - 2 \sin^2 x &= 3 - 2 \frac{(\sqrt{5}-1)^2}{4} \\ &= \frac{12 - 2(5 + 1 - 2\sqrt{5})}{4} \\ &= \frac{12 - 12 + 4\sqrt{5}}{4} = \sqrt{5}\end{aligned}$$

Question10

If $x \neq (2n+1)\frac{\pi}{4}$, then the general solutions of $\cos x + \cos 3x = \sin x + \sin 3x$ is

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Options:

A.

$$n\pi + \frac{\pi}{8}$$

B.

$$n\pi \pm \frac{\pi}{8}$$

C.

$$\frac{n\pi}{2} \pm \frac{\pi}{8}$$

D.

$$\frac{n\pi}{2} + \frac{\pi}{8}$$

Answer: D

Solution:

$$\begin{aligned}\cos x + \cos 3x &= \sin x + \sin 3x \\ \Rightarrow 2 \cos 2x \cdot \cos x &= 2 \sin 2x \cdot \cos x \\ \Rightarrow \cos x (\cos 2x - \sin 2x) &= 0 \\ \text{but given } x &\neq (2n+1)\frac{\pi}{4} \\ \Rightarrow \cos 2x - \sin 2x &= 0 \\ \Rightarrow \sin 2x &= \cos 2x \\ \Rightarrow \tan 2x &= 1 \\ \Rightarrow \tan 2x &= \tan \frac{\pi}{4} \\ \therefore 2x = n\pi + \frac{\pi}{4} &\Rightarrow x = \frac{n\pi}{2} + \frac{\pi}{8}\end{aligned}$$

Question11

The number of solutions of $\sin 2x + \cos 4x = 2$ in the interval $[-\pi, \pi]$ is

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Options:

A.

3

B.

2

C.

0

D.

1

Answer: C

Solution:

$$\sin 2x + \cos 4x = 2x \in [-\pi, \pi]$$

$$\Rightarrow \sin 2x + 1 - 2 \sin^2 2x = 2$$

$$\Rightarrow 2 \sin^2 2x - \sin 2x + 1 = 0$$

$$\Rightarrow \sin 2x = \frac{1 \pm \sqrt{1 - 8}}{2}$$

Not possible

Number of solution = 0

Question12

Number of solutions of the equation

$\cos \theta + \cos 2\theta - \sqrt{3}(\sin \theta + \sin 2\theta) + 1 = 0$ **lying in the interval**
 $(0, 2\pi)$ is

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Options:

A.

3

B.

6

C.

5

D.

4

Answer: D

Solution:

We have,

$$\begin{aligned}\cos \theta + \cos 2\theta - \sqrt{3}(\sin \theta + \sin 2\theta) + 1 &= 0 \\ \Rightarrow \cos \theta + 2\cos^2 \theta - \sqrt{3}\sin \theta - 2\sqrt{3}\sin \theta \cos \theta &= 0 \\ \Rightarrow (\cos \theta - \sqrt{3}\sin \theta) + (2\cos^2 \theta - 2\sqrt{3}\sin \theta \cos \theta) &= 0 \\ \Rightarrow 1(\cos \theta - \sqrt{3}\sin \theta) + 2\cos \theta(\cos \theta - \sqrt{3}\sin \theta) &= 0 \\ \Rightarrow (\cos \theta - \sqrt{3}\sin \theta)(2\cos \theta + 1) &= 0 \\ \Rightarrow \cos \theta - \sqrt{3}\sin \theta = 0 \text{ or } 2\cos \theta = -1 \\ \tan \theta = \frac{1}{\sqrt{3}} \text{ or } \cos \theta = -\frac{1}{2} = \cos \frac{2\pi}{3} \\ \theta = n\pi + \frac{\pi}{6} \text{ or } \theta = 2n\pi + \frac{2\pi}{3} \\ \theta = \frac{\pi}{6}, \frac{7\pi}{6} \text{ or } \theta = \frac{2\pi}{3}, \frac{4\pi}{3}\end{aligned}$$

$\therefore$ Total number of solution in the interval $(0, 2\pi)$ is 4 .

Question13

The number of solutions of the equation

$2\sin^2 \theta - 3\cos^2 \theta = \sin \theta \cos \theta$ lying in the intervals $(-\pi, \pi)$ is

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Options:

A.

2

B.

4

C.

3

D.

1

Answer: B

Solution:

We have,

$$2 \sin^2 \theta - 3 \cos^2 \theta = \sin \theta \cos \theta$$

Dividing both sides by $\cos^2 \theta$, we get $\cos \theta \neq 0$

$$\Rightarrow 2 \tan^2 \theta - 3 = \tan \theta$$

$$\Rightarrow 2 \tan^2 \theta - \tan \theta - 3 = 0$$

$$\Rightarrow (2 \tan \theta - 3)(\tan \theta + 1) = 0$$

$$\therefore \tan \theta = \frac{3}{2} \text{ and } \tan \theta = -1$$

Clearly, in the interval $(-\pi, \pi)$ $\tan \theta = \frac{3}{2}$ gives two solution and $\tan \theta = -1$ gives two solution $\therefore$ Total number of solution = 4

Question14

The general solution of $\cot \frac{x}{2} - \cot x = \operatorname{cosec} \frac{x}{2}$ is

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Options:

A. $\left\{2n\pi + \frac{\pi}{3} \mid n \in \mathbb{Z}\right\}$

B. $\left\{4n\pi + \frac{\pi}{3} \mid n \in \mathbb{Z}\right\}$

C. $\left\{2n\pi + \frac{2\pi}{3} \mid n \in \mathbb{Z}\right\}$

D. $\left\{4n\pi \pm \frac{2\pi}{3} \mid n \in \mathbb{Z}\right\}$

Answer: D

Solution:

Given the equation:

$$\cot \frac{x}{2} - \cot x = \operatorname{cosec} \frac{x}{2}$$

We can express this equation using trigonometric identities:

$$\Rightarrow \frac{\cos \frac{x}{2}}{\sin \frac{x}{2}} - \frac{1}{\sin x} = \operatorname{cosec} \frac{x}{2}$$

$$\Rightarrow \left(\frac{\cos \frac{x}{2} - 1}{\sin \frac{x}{2}} \right) = \frac{\cos x}{\sin x}$$

$$\Rightarrow \frac{\cos \frac{x}{2} - 1}{\sin \frac{x}{2}} = \frac{\cos x}{2 \sin \frac{x}{2} \cos \frac{x}{2}}$$

$$\Rightarrow 2(\cos \frac{x}{2} - 1) = \frac{\cos x}{\cos \frac{x}{2}}$$

$$\Rightarrow 2 \cos^2 \frac{x}{2} - 2 \cos \frac{x}{2} = \cos x$$

Using the identity $\cos x = 2 \cos^2 \frac{x}{2} - 1$:

$$\Rightarrow 2 \cos^2 \frac{x}{2} - 2 \cos \frac{x}{2} = 2 \cos^2 \frac{x}{2} - 1$$

$$\Rightarrow \cos \frac{x}{2} = \frac{1}{2} = \cos \frac{\pi}{3}$$

This implies:

$$\Rightarrow \frac{x}{2} = 2n\pi \pm \frac{\pi}{3}$$

Thus, we find the general solution for x :

$$\Rightarrow x = 4n\pi \pm \frac{2\pi}{3}$$

Question15

The values of x in $(-\pi, \pi)$, which satisfy the equation $8^{1+\cos^2 x + \cos^4 x + \dots} = 4^3$ are

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Options:

A. $\pm \frac{\pi}{4}, \pm \frac{3\pi}{4}$

B. $\pm \frac{\pi}{6}, \frac{\pi}{3}$

C. $\pm \frac{\pi}{8}$

D. $\frac{\pi}{3}$

Answer: A

Solution:

Given that

$$8^{1+\cos^2 x + \cos^4 x + \dots} = 4^3$$

$$\Rightarrow 2^{3(1+\cos^2 x + \cos^4 x + \dots)} = 2^6$$

$$\Rightarrow 3(1 + \cos^2 x + \cos^4 x + \dots) = 6$$

$$\Rightarrow 1 + \cos^2 x + \cos^4 x + \dots = 2$$

$$\Rightarrow \cos^2 x + \cos^4 x + \dots = 1$$

$$\therefore \text{Sum of GP of infinite terms} = \frac{a}{1-r}$$

$$\therefore \frac{\cos^2 x}{1-\cos^2 x} = 1 \Rightarrow \cos^2 x = 1 - \cos^2 x$$

$$\Rightarrow 2 \cos^2 x = 1 \Rightarrow \cos^2 x = \frac{1}{2}$$

$$\Rightarrow x = \pm \frac{\pi}{4}, \pm \frac{3\pi}{4}$$

Question 16

The general solution of the equation $\tan x + \tan 2x - \tan 3x = 0$ is

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Options:

A. $\{x \mid x = n\pi \pm \frac{\pi}{3} \text{ or } \frac{n\pi}{2}, n \in \mathbb{Z}\}$

B. $\{x \mid x = n\pi \pm \frac{\pi}{3} \text{ or } n\pi, n \in \mathbb{Z}\}$

C. $\{x \mid x = n\pi \pm \frac{\pi}{3} \text{ or } \frac{n\pi}{2} \text{ or } n\pi, n \in \mathbb{Z}\}$

D. $\{x \mid x = n\pi \pm \frac{\pi}{6} \text{ or } \frac{n\pi}{2}, n \in \mathbb{Z}\}$

Answer: B

Solution:

To solve the equation $\tan x + \tan 2x - \tan 3x = 0$, we start by understanding the identity given:

$$\tan x + \tan 2x = \tan 3x.$$

Using trigonometric identities, we have:

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

$$\tan 3x = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}.$$

Substituting these into the equation gives:

$$\tan x + \frac{2 \tan x}{1 - \tan^2 x} = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}.$$

Simplifying this, we obtain:

$$\frac{\tan x(3-\tan^2 x)}{1-\tan^2 x} = \frac{\tan x(3-\tan^2 x)}{1-3\tan^2 x}.$$

Clearing the denominators by multiplying through by $(1-\tan^2 x)(1-3\tan^2 x)$, we get:

$$\tan x(3-\tan^2 x)(1-3\tan^2 x) = \tan x(3-\tan^2 x)(1-\tan^2 x).$$

Subtracting gives:

$$\tan x(3-\tan^2 x)\{-2\tan^2 x\} = 0.$$

This implies:

$$\tan^3 x = 0 \text{ or}$$

$$3-\tan^2 x = 0.$$

Solving each equation separately:

$$\tan^3 x = 0 \implies \tan x = 0 \implies x = n\pi, n \in \mathbb{Z}.$$

$$3-\tan^2 x = 0 \implies \tan^2 x = 3 \implies \tan x = \pm\sqrt{3}.$$

Solving $\tan x = \pm\sqrt{3}$, we get:

$$\tan x = \sqrt{3} \implies x = n\pi + \frac{\pi}{3}, n \in \mathbb{Z}.$$

$$\tan x = -\sqrt{3} \implies x = n\pi - \frac{\pi}{3}, n \in \mathbb{Z}.$$

Therefore, the general solution is:

$$\{x \mid x = n\pi \pm \frac{\pi}{3}, n\pi, n \in \mathbb{Z}\}.$$

Question17

Number of solutions of the trigonometric equation $2 \tan 2\theta - \cot 2\theta + 1 = 0$ lying in the interval $[0, \pi]$ is

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Options:

A. 2

B. 3

C. 4

D. 5

Answer: C

Solution:

Given, $2 \tan 2\theta - \cot 2\theta + 1 = 0, \theta \in [0, x]$

$$\Rightarrow 2 \tan 2\theta - \frac{1}{\tan 2\theta} + 1 = 0$$

$$\Rightarrow 2 \tan^2 2\theta + \tan 2\theta - 1 = 0$$

$$\Rightarrow 2 \tan^2 2\theta + 2 \tan 2\theta - \tan 2\theta - 1 = 0$$

$$\Rightarrow 2 \tan 2\theta(1 + \tan 2\theta) - (1 + \tan 2\theta) = 0$$

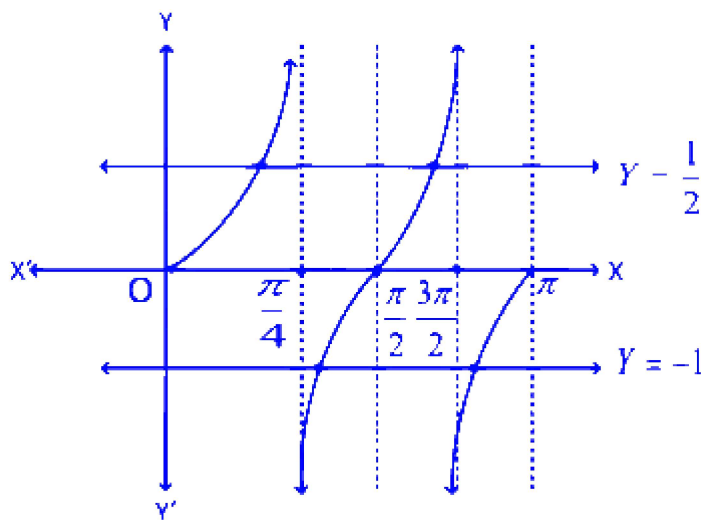
$$\Rightarrow (2 \tan 2\theta - 1)(1 + \tan 2\theta) = 0$$

$$\Rightarrow 2 \tan 2\theta = 1 \text{ or } \tan 2\theta = -1$$

$$\Rightarrow \tan 2\theta = \frac{1}{2}$$

$$\text{or } \tan 2\theta = \tan \left(n\pi + \frac{3\pi}{4} \right)$$

$$\Rightarrow 2\theta = n\pi + \frac{3\pi}{4} \Rightarrow \theta = \frac{n\pi}{2} + \frac{3\pi}{8}$$



Since, number of point of intersection is 4 .

Hence, number of solution is 4 .

Question18

For $a \in \mathbb{R} - \{0\}$, if $a \cos x + a \sin x + a = 2k + 1$ has a solution, then k lies in the interval

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Options:

$$\text{A. } \left[\frac{a-1-\sqrt{2}a}{2}, \frac{a-1+\sqrt{2}a}{2} \right]$$

$$\text{B. } \left[\frac{a+1-\sqrt{2}}{2}, \frac{a+1+\sqrt{2}}{2} \right]$$

$$\text{C. } \left[\frac{a-1-\sqrt{2}}{2}, \frac{a-1+\sqrt{2}}{2} \right]$$

$$\text{D. } \left[-\frac{(\sqrt{2a^2+2a+1}+1)}{2}, \frac{(\sqrt{2a^2+2a+1}-1)}{2} \right]$$

Answer: A

Solution:

Given the equation:

$$a \cos x + a \sin x + a = 2k + 1$$

we can rearrange it as:

$$a(\cos x + \sin x) = 2k + 1 - a$$

Divide through by a (since $a \neq 0$):

$$\cos x + \sin x = \frac{2k+1-a}{a} = \frac{2k+1}{a} - 1$$

Recognizing that:

$$\cos x + \sin x = \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos x + \frac{1}{\sqrt{2}} \sin x \right)$$

Let $\frac{1}{\sqrt{2}} \cos x + \frac{1}{\sqrt{2}} \sin x = \sin \left(x + \frac{\pi}{4} \right)$, then:

$$\sqrt{2} \sin \left(x + \frac{\pi}{4} \right) = \frac{2k+1}{a} - 1$$

So:

$$\sin \left(x + \frac{\pi}{4} \right) = \frac{1}{\sqrt{2}} \left(\frac{2k+1-a}{a} \right)$$

For $\sin(y)$ where y is in the range $[-1, 1]$:

$$-1 \leq \frac{2k+1-a}{\sqrt{2}a} \leq 1$$

This implies:

$$-a\sqrt{2} \leq 2k + 1 - a \leq a\sqrt{2}$$

Rearranging gives:

$$a - a\sqrt{2} \leq 2k + 1 \leq a\sqrt{2} + a$$

Subtract 1 from each part:

$$a - a\sqrt{2} - 1 \leq 2k \leq a\sqrt{2} + a - 1$$

Finally, divide by 2:

$$\frac{a-a\sqrt{2}-1}{2} \leq k \leq \frac{a\sqrt{2}+a-1}{2}$$

Thus, k lies in the interval:

$$\left[\frac{a-a\sqrt{2}-1}{2}, \frac{a\sqrt{2}+a-1}{2} \right]$$

Therefore, k is within the interval $\left[\frac{a-1-\sqrt{2}a}{2}, \frac{a-1+\sqrt{2}a}{2} \right]$.

Question19

If the general solution .set of $\sin x + 3 \sin 3x + \sin 5x = 1$ is S , then $\{\sin \alpha / \alpha \in S\} =$

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Options:

- A. $\{1, -1, 0\}$
- B. $\{\frac{1}{2}, \frac{-1}{2}, 0, 1, -1\}$
- C. $\{\frac{\sqrt{3}}{2}, 0, \frac{-\sqrt{3}}{2}\}$
- D. $\{1, -1, \frac{\sqrt{3}}{2}, 0, \frac{-\sqrt{3}}{2}\}$

Answer: C

Solution:

To solve the equation $\sin x + 3 \sin 3x + \sin 5x = 1$, we start by rewriting it as:

$$(\sin x + \sin 5x) + 3 \sin 3x = 0$$

Using the sum-to-product identities, the expression $\sin x + \sin 5x$ becomes:

$$2 \sin 3x \cos 2x$$

Thus, the equation becomes:

$$2 \sin 3x \cos 2x + 3 \sin 3x = 0$$

Factoring out $\sin 3x$, we get:

$$\sin 3x(2 \cos 2x + 3) = 0$$

Which implies that:

$$\sin 3x = 0$$

$$2 \cos 2x + 3 = 0$$

For $\sin 3x = 0$, the solutions are:

$$3x = n\pi \Rightarrow x = \frac{n\pi}{3}, \quad n \in \mathbb{Z}$$

Next, for $2 \cos 2x + 3 = 0$, solving gives:

$$\cos 2x = -\frac{3}{2}$$

However, since the cosine of an angle cannot exceed 1 or be less than -1, there are no solutions in real numbers for this equation.

Since the only valid solutions come from $\sin 3x = 0$, we consider the possible values of $\sin \alpha$ where $\alpha \in S$. Evaluating $\sin$ at these angles, we find:

$$\{\sin \alpha \mid \alpha \in S\} = \left\{ \frac{\sqrt{3}}{2}, 0, \frac{-\sqrt{3}}{2} \right\}$$

Question20

The general solution of the equation $\sin^2 \theta + 3 \cos^2 \theta = 5 \sin \theta$ is

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Options:

A. $n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$

B. $n\pi + (-1)^n \frac{\pi}{6}, n \in \mathbb{Z}$

C. $n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$

D. $n\pi + (-1)^n \frac{\pi}{3}, n \in \mathbb{Z}$

Answer: B

Solution:

Given, $\sin^2 \theta + 3 \cos^2 \theta = 5 \sin \theta$

$$\Rightarrow \sin^2 \theta + 3(1 - \sin^2 \theta) = 5 \sin \theta$$

$$\Rightarrow \sin^2 \theta + 3 - 3 \sin^2 \theta = 5 \sin \theta$$

$$\Rightarrow -2 \sin^2 \theta - 5 \sin \theta + 3 = 0$$

$$\Rightarrow 2 \sin^2 \theta + 5 \sin \theta - 3 = 0$$

$$\Rightarrow \sin \theta = \frac{-6}{2}, \sin \theta = \frac{1}{2}$$

$$\therefore \sin \theta = \frac{1}{2} \quad [\because |\sin \theta| \leq 1]$$

$$\Rightarrow \theta = n\pi + (-1)^n \frac{\pi}{6}$$

$\therefore$ The general solution is $n\pi + (-1)^n \frac{\pi}{6}, n \in \mathbb{Z}$.

Question21

The number of ordered pairs (x, y) satisfying the equations $\sin x + \sin y = \sin(x + y)$ and $|x| + |y| = 1$ is

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Options:

A. 2

B. 3

C. 4

D. 6

Answer: D

Solution:

We are given two equations: $\sin x + \sin y = \sin(x + y)$ and $|x| + |y| = 1$.

Let's analyze the first equation:

$$\sin x + \sin y = \sin(x + y)$$

$$\Rightarrow 2 \sin \left(\frac{x+y}{2} \right) \cdot \cos \left(\frac{x-y}{2} \right) = 2 \sin \left(\frac{x+y}{2} \right) \cdot \cos \left(\frac{x+y}{2} \right)$$

$$\Rightarrow \sin \left(\frac{x+y}{2} \right) \left[\cos \left(\frac{x-y}{2} \right) - \cos \left(\frac{x+y}{2} \right) \right] = 0$$

$$\Rightarrow \sin \left(\frac{x+y}{2} \right) \left[-2 \sin \frac{x}{2} \cdot \sin \left(-\frac{y}{2} \right) \right] = 0$$

$$\Rightarrow 2 \sin \left(\frac{x+y}{2} \right) \sin \frac{x}{2} \cdot \sin \frac{y}{2} = 0$$

From this, we have three cases to consider:

Case 1: $\sin\left(\frac{x+y}{2}\right) = 0$

This implies:

$$\frac{x+y}{2} = 0 \Rightarrow x + y = 0 \Rightarrow x = -y$$

Using the second equation $|x| + |y| = 1$:

$$|x| + |-x| = 1 \Rightarrow x = \pm \frac{1}{2} \quad \text{and} \quad y = \mp \frac{1}{2}$$

Thus, the ordered pairs are $\left(\frac{1}{2}, -\frac{1}{2}\right)$ and $\left(-\frac{1}{2}, \frac{1}{2}\right)$.

Case 2: $\sin \frac{x}{2} = 0$

This gives:

$$\frac{x}{2} = 0 \Rightarrow x = 0$$

From $|x| + |y| = 1$:

$$0 + |y| = 1 \Rightarrow |y| = 1 \Rightarrow y = \pm 1$$

So, the ordered pairs are $(0, 1)$ and $(0, -1)$.

Case 3: $\sin \frac{y}{2} = 0$

This implies:

$$\frac{y}{2} = 0 \Rightarrow y = 0$$

Using $|x| + |y| = 1$:

$$|x| + 0 = 1 \Rightarrow |x| = 1 \Rightarrow x = \pm 1$$

Therefore, the ordered pairs are $(1, 0)$ and $(-1, 0)$.

By combining all the cases, we have a total of 6 ordered pairs:

$$\left(\frac{1}{2}, -\frac{1}{2}\right)$$

$$\left(-\frac{1}{2}, \frac{1}{2}\right)$$

$$(0, 1)$$

$$(0, -1)$$

$$(1, 0)$$

$$(-1, 0)$$

Question22

If $5 \sinh hx - \cosh hx = 5$, then one of the values of $\tanh hx$ is

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Options:

A. $\frac{2}{5}$

B. $\frac{3}{5}$

C. $\frac{-3}{5}$

D. $\frac{-1}{5}$

Answer: C

Solution:

We have, $5 \sinh(x) - \cosh(x) = 5$

$$\Rightarrow 5 \left(\frac{e^x - e^{-x}}{2} \right) - \left(\frac{e^x + e^{-x}}{2} \right) = 5$$

$$\Rightarrow \frac{5e^x - 5e^{-x} - e^x - e^{-x}}{2} = 5$$

$$\Rightarrow \frac{4e^x - 6e^{-x}}{2} = 5 \Rightarrow 2e^x - 3e^{-x} = 5$$

$$\Rightarrow 2e^{2x} - 3 = 5e^x \Rightarrow 2e^{2x} - 5e^x - 3 = 0$$

Let, $e^x = y$

Then, $2y^2 - 5y - 3 = 0$

$$\begin{aligned} y &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{5 \pm \sqrt{(-5)^2 - 4 \times 2 \times (-3)}}{2 \times 2} \\ &= \frac{5 \pm \sqrt{25 + 24}}{4} = \frac{5 \pm \sqrt{49}}{4} = \frac{5 \pm 7}{4} \\ &= \frac{5+4}{4}, \frac{5-7}{4} = 3, -\frac{1}{2} \end{aligned}$$

Thus, $e^x = 3$ or $e^x = -\frac{1}{2}$

If $e^x = 3$

Then,

$$\begin{aligned}\tanh x &= \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{3 - \frac{1}{3}}{3 + \frac{1}{3}} \\ &= \frac{8/3}{10/3} = \frac{8}{10} = \frac{4}{5}\end{aligned}$$

If $e^x = -\frac{1}{2}$

$$\begin{aligned}\text{Then, } \tanh(x) &= \frac{e^x - e^{-x}}{e^x + e^{-x}} \\ &= \frac{-\frac{1}{2} + 2}{-\frac{1}{2} - 2} = \frac{\frac{3}{2}}{-\frac{5}{2}} = -\frac{3}{5}\end{aligned}$$

Question23

The smallest positive value (in degrees) of θ for which $\tan(\theta + 100^\circ) = \tan(\theta + 50^\circ) \tan(\theta) \tan(\theta - 50^\circ)$ is valid, is

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Options:

A. 60°

B. 45°

C. 30°

D. 15°

Answer: C

Solution:

We need to find the smallest positive value of θ for which the equation

$$\tan(\theta + 100^\circ) = \tan(\theta + 50^\circ) \tan(\theta) \tan(\theta - 50^\circ)$$

holds true.

Step-by-Step Solution:

Start with the given equation:

$$\tan(\theta + 100^\circ) = \tan(\theta + 50^\circ) \tan(\theta) \tan(\theta - 50^\circ)$$

Rearranging terms, we have:

$$\frac{\tan(\theta+100^\circ)}{\tan(\theta-50^\circ)} = \tan(\theta+50^\circ)\tan(\theta)$$

Express tangent in terms of sine and cosine:

$$\frac{\sin(\theta+50^\circ)}{\cos(\theta+100^\circ)\cos(\theta-50^\circ)} = \frac{\sin(\theta+50^\circ)\sin(\theta-50^\circ)}{\cos(\theta+50^\circ)\cos(\theta)}$$

Using the componendo and dividendo rule:

$$\frac{\sin(2\theta+50^\circ)}{\sin(150^\circ)} = \frac{-\cos 50^\circ}{\cos(2\theta+50^\circ)}$$

Equating the terms, we proceed with the identities:

$$\sin(2\theta+50^\circ)\cos(2\theta+50^\circ) = -\cos 50^\circ\sin 150^\circ$$

Multiply both sides by 2:

$$\sin(4\theta+100^\circ) = -\sin 40^\circ$$

Since $\sin(-x) = -\sin(x)$, we have:

$$\sin(4\theta+100^\circ) = \sin(-40^\circ)$$

The general solution for the sine equation is:

$$4\theta+100^\circ = n \cdot 180^\circ + (-1)^n \cdot (-40^\circ)$$

For $n = 1$, substitute:

$$4\theta+100^\circ = 180^\circ + 40^\circ$$

$$4\theta+100^\circ = 220^\circ$$

Solve for θ :

$$4\theta = 220^\circ - 100^\circ$$

$$4\theta = 120^\circ$$

$$\theta = 30^\circ$$

Thus, the smallest positive value of θ is $\boxed{30^\circ}$.

Question 24

The general solution of

$$4 \cos 2x - 4\sqrt{3} \sin 2x + \cos 3x - \sqrt{3} \sin 3x \\ + \cos x - \sqrt{3} \sin x = 0$$

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Options:

A. $\frac{n\pi}{2} - \frac{\pi}{3}$

B. $\frac{n\pi}{2} + \frac{\pi}{6}$

C. $\frac{n\pi}{2} + \frac{\pi}{12}$

D. $\frac{n\pi}{2} - \frac{\pi}{12}$

Answer: C

Solution:

$$4 \cos 2x - 4\sqrt{3} \sin 2x + \cos 3x$$

$$-\sqrt{3} \sin 3x + \cos x - \sqrt{3} \sin x = 0$$

Divide by 2 on both sides, we get

$$\left[\frac{1}{2} \cos 2x - \frac{\sqrt{3}}{2} \sin 2x \right] + \left[\frac{1}{2} \cos 3x - \frac{\sqrt{3}}{2} \sin 3x \right] + \left[\frac{1}{2} \cos x - \frac{\sqrt{3}}{2} \sin x \right] = 0 \Rightarrow \left[\sin \frac{\pi}{6} \cos 2x - \cos \frac{\pi}{6} \sin 2x \right]$$

$$+ \left[\sin \frac{\pi}{6} \cos 3x - \cos \frac{\pi}{6} \sin 3x \right] + \left[\sin \frac{\pi}{6} \cos x - \cos \frac{\pi}{6} \sin x \right] = 0$$

$$\Rightarrow 4 \sin \left(\frac{\pi}{6} - 2x \right) + \sin \left(\frac{\pi}{6} - 3x \right)$$

$$+ \sin \left(\frac{\pi}{6} - x \right) = 0$$

$$\Rightarrow 4 \sin \left(\frac{\pi}{6} - 2x \right) + 2 \sin \left(\frac{\pi}{6} - 2x \right)$$

$$\Rightarrow$$

$$\Rightarrow 2 \sin \left(\frac{\pi}{6} - 2x \right) [\cos x + 2] = 0$$

$$\Rightarrow 2 \sin \left(\frac{\pi}{6} - 2x \right) = 0$$

$$\frac{\pi}{6} - 2x = -n\pi$$

$$2x = n\pi + \frac{\cos x \neq -2}{6} \Rightarrow x = \frac{n\pi}{2} + \frac{\pi}{12}$$

Question25

The general solution of $2 \cos^2 x - 2 \tan x + 1 = 0$ is

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Options:

A. $n\pi + \frac{\pi}{4}, n \in \mathbb{Z}$

B. $2n\pi \pm \frac{\pi}{4}, n \in \mathbb{Z}$

C. $2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$

D. $n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$

Answer: A

Solution:

To solve the equation $2 \cos^2 x - 2 \tan x + 1 = 0$, we proceed with the following steps:

Start by rewriting the equation:

$$2 \cos^2 x - 2 \tan x + 1 = 0$$

Using the identity $\cos^2 x = \frac{1+\cos 2x}{2}$, substitute:

$$2 \left(\frac{1+\cos 2x}{2} \right) - 2 \tan x + 1 = 0$$

This simplifies to:

$$\cos 2x - 2 \tan x + 2 = 0$$

Express $\cos 2x$ in terms of $\tan x$:

$$\cos 2x = \frac{1-\tan^2 x}{1+\tan^2 x}$$

Substitute this identity:

$$\frac{1-\tan^2 x}{1+\tan^2 x} - 2 \tan x + 2 = 0$$

Eliminate the fraction by multiplying through by $1 + \tan^2 x$:

$$1 - \tan^2 x - 2 \tan x + 2 \tan^2 x + 2 = 0$$

Simplify the equation:

$$-2 \tan^3 x + \tan^2 x - 2 \tan x + 3 = 0$$

Rearrange and factor the polynomial, letting $\tan x = y$:

$$2y^3 - y^2 + 2y - 3 = 0$$

Factoring, we get:

$$(y - 1)(2y^2 + y + 3) = 0$$

Solve the factors:

$$y - 1 = 0 \Rightarrow y = 1$$

The quadratic $2y^2 + y + 3$ does not have real solutions (since its discriminant is negative).

Therefore, $\tan x = 1$, which implies:

$$x = n\pi + \frac{\pi}{4}, \quad n \in \mathbb{Z}$$

The general solution to the equation is thus $x = n\pi + \frac{\pi}{4}$ where n is any integer.

Question26

$$1 + \sin x + \sin^2 x + \sin^3 x + \dots + \infty = 4 + 2\sqrt{3} \text{ and } 0 < x < \pi, x \neq \frac{\pi}{2} \text{ then } x =$$

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Options:

A.

$$\frac{\pi}{6}, \frac{\pi}{4}$$

B.

$$\frac{\pi}{4}, \frac{5\pi}{6}$$

C.

$$\frac{2\pi}{5}, \frac{\pi}{6}$$

D.

$$\frac{\pi}{3}, \frac{2\pi}{3}$$

Answer: D

Solution:

To solve the given series equation:

$$1 + \sin x + \sin^2 x + \sin^3 x + \dots = 4 + 2\sqrt{3}$$

we recognize that this is an infinite geometric series with the first term $a = 1$ and the common ratio $r = \sin x$. The sum of an infinite geometric series is given by:

$$S = \frac{a}{1-r}$$

Here, substituting the given values:

$$\frac{1}{1-\sin x} = 4 + 2\sqrt{3}$$

To find $\sin x$, we solve:

$$1 - \sin x = \frac{1}{4+2\sqrt{3}}$$

Rationalizing the denominator gives:

$$1 - \sin x = \frac{4-2\sqrt{3}}{4} = 1 - \frac{\sqrt{3}}{2}$$

Thus:

$$\sin x = \frac{\sqrt{3}}{2}$$

Given the constraints $0 < x < \pi$ and $x \neq \frac{\pi}{2}$, we find that x could be $\frac{\pi}{3}$ or $\frac{2\pi}{3}$. The solutions within this range are:

$$x = \frac{\pi}{3}, \frac{2\pi}{3}$$

Question27

If $\sin \theta + \operatorname{cosec} \theta = 4$, then $\sin^2 \theta + \operatorname{cosec}^2 \theta =$

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Options:

A. 12

B. 18

C. 16

D. 14

Answer: D

Solution:

We have, $\sin \theta + \operatorname{cosec} \theta = 4$

$$\Rightarrow \frac{1}{\operatorname{cosec} \theta} + \operatorname{cosec} \theta = 4$$

Squaring both sides,

$$\begin{aligned}\frac{1}{\operatorname{cosec}^2 \theta} + \operatorname{cosec}^2 \theta + 2 &= 16 \\ \Rightarrow \sin^2 \theta + \operatorname{cosec}^2 \theta &= 16 - 2 = 14\end{aligned}$$

Question28

If $2 \cosh 2x + 10 \sinh 2x = 5$, then $x =$

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Options:

A. $\frac{1}{2} \log \frac{4}{3}$

B. $\frac{1}{2} \log \frac{2}{3}$

C. $\frac{1}{2} \log \frac{3}{2}$

D. $\frac{1}{2} \log \frac{3}{4}$

Answer: A

Solution:

Here, $2 \cosh 2x + 10 \sinh 2x = 5$

We know that,

$$\cosh 2x = \frac{1}{2}(e^{2x} + e^{-2x})$$

$$\sinh 2x = \frac{1}{2}(e^{2x} - e^{-2x})$$

$$\text{So, } 2 \left[\frac{1}{2}(e^{2x} + e^{-2x}) \right] + 10 \left[\frac{1}{2}(e^{2x} - e^{-2x}) \right] = 5$$

$$\Rightarrow e^{2x} + e^{-2x} + 5e^{2x} - 5e^{-2x} = 5$$

$$\Rightarrow 6e^{2x} - 5 - 4e^{-2x} = 0 \Rightarrow (3e^{2x} - 4)(2e^{2x} + 1) = 0$$

$$e^{2x} = \frac{4}{3} \text{ or } e^{2x} = -\frac{1}{2}$$

The only real solution occurs when $e^{2x} > 0$.

$$\text{So, } 2x = \ln \frac{4}{3}$$

$$\Rightarrow x = \frac{1}{2} \ln \frac{4}{3}$$

Question29

If $\sin \left(\frac{\pi}{4} \cos \theta \right) = \cos \left(\frac{\pi}{4} \tan \theta \right)$, then θ is equal to

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Options:

A. $2n\pi + \frac{\pi}{4}$

B. $2n\pi \pm \frac{\pi}{4}$

C. $2n\pi - \frac{\pi}{4}$

D. $n\pi + \frac{\pi}{4}$

Answer: D

Solution:

$$\sin\left(\frac{\pi}{4}\cot\theta\right) = \cos\left(\frac{\pi}{4}\tan\theta\right)$$

$$\sin A = \cos B$$

$$\text{If } A + B = \frac{\pi}{2}$$

$$\Rightarrow \sin A = \cos B$$

$$\Rightarrow \frac{\pi}{4}(\tan\theta + \cot\theta) = \frac{\pi}{2}$$

$$\Rightarrow \tan\theta + \frac{1}{\tan\theta} = 2 \Rightarrow \tan^2\theta \cdot 2\tan\theta + 1 = 0$$

$$\Rightarrow (\tan\theta - 1)^2 = 0 \Rightarrow \tan^2\theta - 1 = 0$$

$$\Rightarrow \tan\theta = 1$$

$$\theta = n\pi + \frac{\pi}{4}$$

Question30

If $\theta \in [0, 2\pi]$ and $\cos 2\theta = \cos \theta + \sin \theta$, then the sum of all values of θ satisfying the equation is

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Options:

A. $\frac{21\pi}{2}$

B. $\frac{11\pi}{4}$

C. $\frac{24\pi}{4}$

D. $\frac{31\pi}{4}$

Answer: C

Solution:

$$\cos 2\theta = \cos \theta + \sin \theta$$

$$\Rightarrow (\cos^2 \theta - \sin^2 \theta) - (\cos \theta + \sin \theta) = 0$$

$$\Rightarrow (\cos \theta + \sin \theta)(\cos \theta - \sin \theta - 1) = 0$$

$$\Rightarrow \cos \theta + \sin \theta = 0$$

$$\text{or } \cos \theta - \sin \theta = 1$$

When $\cos \theta + \sin \theta = 0$, then $\tan \theta = -1$

$$\text{then } \theta = n\pi + \frac{3\pi}{4} \Rightarrow \theta = \frac{3\pi}{4}, \frac{7\pi}{4} \in [0, 2\pi]$$

and when $\cos \theta - \sin \theta = 1$, then $\sin \theta + (1 - \cos \theta) = 0$

$$\text{and } 2 \sin \frac{\theta}{2} \left(\cos \frac{\theta}{2} + \sin \frac{\theta}{2} \right) = 0$$

$$\text{and } \sin \frac{\theta}{2} = 0, \cos \frac{\theta}{2} + \sin \frac{\theta}{2} = 0$$

$$\Rightarrow \frac{\theta}{2} = n\pi, \tan \frac{\theta}{2} = -1$$

$$\Rightarrow \theta = 0, 2\pi \in [0, 2\pi],$$

$$\theta = \frac{6\pi}{4} \in [0, 2\pi]$$

Sum of all values of θ

$$= \frac{3\pi}{4} + \frac{7\pi}{4} + 0 + 2\pi + \frac{6\pi}{4} = \frac{24\pi}{4}$$

Question31

The value of x satisfying the equation $3 \operatorname{cosec} x = 4 \sin x$ are

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Options:

A. $\frac{\pi}{6}, \frac{\pi}{3}$

B. $\pm \frac{\pi}{6}$

C. $\pm \frac{\pi}{3}$

D. $\frac{\pi}{3}, \frac{\pi}{4}$

Answer: C

Solution:

$$3 \operatorname{cosec} x = 4 \sin x$$

$$\text{or } \sin^2 x = 3/4$$

$$\text{or } \sin^2 x = \sin^2 \frac{\pi}{3} \Rightarrow x = n\pi \pm \frac{\pi}{3}, n \in Z$$

$$\text{or } x = \pm \frac{\pi}{3}$$
